

Shabab Ahmed

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PROFESSIONAL PROFILE

Ph.D. Candidate in Economics specializing in contract theory and structural modeling. Applied the latter during a pricing-focused economics internship at Amazon.

EDUCATION

Ph.D. in Economics, University of Washington

Expected December 2026

Dissertation:

- [Dynamic Moral Hazard under Preference Change](#): how anticipated changes in future preferences affect incentives today, applied to the NFL draft policy. Currently testing the model's predictions using quasi-experimental methods.
- [High-Dimensional Demand Estimation](#): applies Lasso and Elastic Net regularization with debiasing corrections to estimate elasticity under collinearity with potentially many instruments and regressors. Validated via Monte Carlo simulation.
- [Fixed Budget Allocation under Hidden Effort](#): how agents spend across time under a budget constraint. This offers a new explanation for observed patterns like budget hoarding and end-of-year spending spikes.

Passed general examination; presented research in departmental seminars.

M.A. in Economics, University of Washington

Dec 2022 — GPA 3.89

M.S. in Applied Mathematics, University of Washington

Jun 2020 — GPA 3.93

B.A. in Mathematics and Economics, Colby College

May 2019 — GPA 3.93

Magna Cum Laude, Phi Beta Kappa, Distinction in Economics & Mathematics, Dean's List (6 semesters).

EXPERIENCE

Economist Intern, Amazon, Bellevue, WA

Jul 2025 – Oct 2025

- Built structural demand models on large-scale survey data to estimate customer preferences for Project Kuiper. Authored and presented a technical and business report to senior leadership across multiple teams, who evaluated it as an input to pricing decisions.
- Built a Python tool that returned projected market share and revenue based on user-entered pricing and product configurations. The team used it to run counterfactual business scenarios.

Graduate Teaching Assistant & Independent Instructor, University of Washington

2019 – Present

- Independent Instructor for Game Theory and Managerial Economics, with full responsibility for course delivery, assignments, and assessment.
- Teaching Assistant for Graduate Microeconomic Theory and undergraduate courses, leading discussion sections and review sessions.
- Summer Study Section Instructor, Math & Science Upward Bound (Summers 2024 and 2026): mentored high school students on independent research projects.

EARLIER PROJECTS

[NBA Decade Comparison: PCA and Machine Learning Analysis](#)

Winter 2020

- Partnered with a classmate to build classifiers on 13,000 player-season observations (~95 features scraped from Basketball Reference). Used PCA and LDA to predict player position and decade. Used L1-regularized logistic regression to predict Win Shares quartile.
- Achieved 72.6% accuracy predicting Win Shares quartile for 1990s-decade players, cross-validated over 50 iterations.

[Game Theory in Cricket](#)

Spring 2019

- Built a game-theoretic model of bowler-batsman strategy and tested Nash equilibrium predictions against a hand-coded dataset of bowler-batsman interactions from 2017 ICC Champions Trophy match commentary.
- Found predicted strategy frequencies matched actual play within one bootstrapped standard deviation under a simplified model, though the result did not hold under more granular or alternate specifications.

TECHNICAL SKILLS

Causal Inference & Econometrics: IV, DiD/event study

Structural & Statistical Methods: structural demand estimation, dynamic optimization, linear and logistic regression, Lasso/Elastic Net regularization, debiased ML, PCA, LDA, bootstrapping, cross-validation, Monte Carlo simulation

Languages & Tools: Python, R, Stata, SQL, MATLAB, Sage, Mathematica, LaTeX, Qualtrics, Excel

Languages: English (fluent), Bengali (fluent), Hindi (conversational)